

ANSWERS

1.

Question	Answer	Marks
4(a)(i)	13 A	B1
4(a)(ii)	0 (A)	B1
4(b)(i)	resistance of earth wire (and casing) small or disconnects circuit or no electric shock (possible) or stops current	B1
	current in (live and) earth wire (briefly)	B1
	fuse (in live wire) blows / melts	B1
4(b)(ii)	voltage (source) disconnected (by fuse's blowing) or live voltage immediately next to (switch and blown) fuse or it / live wire at high / maximum voltage / potential	B1
	no part of the washing machine / casing is live or no current to <u>casing</u>	B1

2.

Question	Answer	Marks
9(a)	sinusoidal voltage-time graph labelled a.c. drawn	B1
	voltage-time labelled d.c. graph with horizontal, non-zero line	B1
	an a.c. varies with time or an a.c. changes direction / polarity (at regular time intervals) or a d.c. is constant with time or d.c. is never negative	B1
9(b)(i)	$(I =) P / V$ or 60 / 240	C1
	0.25 A	A1
9(b)(ii)	$(R =) V / I$ or 240 / 0.25	C1
	960 Ω	A1
9(b)(iii)	any two from: if one lamp blows the others stay lit can be operated separately each lamp has the correct voltage across it or lamps glow brightly / normally	B2
9(b)(iv)	(number of units =) $5 \times 60 \times 5.5 \times 365 / 1000$ or 600 or 602.25 or 18(.07)	C1
	\$90 or \$90.34	A1
9(c)(i)	cell, lamp and ammeter in series and no other components	B1
	voltmeter in parallel with lamp	B1
9(c)(ii)	(in this circuit) the resistance is (very much) smaller	B1
	(metal) filament is cooler	B1

3.

Question	Answer	Marks
5(a)	if the live / high voltage wire touches the casing / machine or if the casing / machine becomes live	B1
	large current in earth wire or resistance of earth wire small	B1
	fuse melts or disconnects supply	B1
5(b)(i)	live wire	B1
5(b)(ii)	when the fuse melts the live wire is not connected to any part of the device /is isolated from the mains / supply	B1
5(c)(i)	live and neutral	B1
5(c)(ii)	double insulation or it / casing / hair dryer is plastic	B1

4.

Question	Answer	Marks
6(a)	worn insulation → visual check loose wire in live → earth wire cable too hot → circuit breaker ALL three correct B2 one or two correct B1	B2
6(b)(i)	energy used by (a device of power) 1 kW for 1 hour	B1
6(b)(ii)	(number of kW h =) $1.2 \times 20 / 60$ or 0.4 seen	C1
	8 c	A1

5.

Question	Answer	Marks
8(a)	any 2 from - earthing - contact breaker / circuit breaker - double insulation	B2
8(b)(i)	fuses are in live wire or fuse is connected to W / it	B1
8(b)(ii)	switch drawn between the letter W on the diagram and the left-hand lamp	B1
8(b)(iii)	current is larger than 5 A fuse blows / melts / breaks (circuit) or can only pass currents up to 5 A	B1

6.

Question	Answer	Marks
6(a)(i)	200 W	B1
6(a)(ii)	power $\times$ time or any numerical value of power $\times$ 3	C1
	0.6(0)	A1
6(a)(iii)	(I =) P / V algebraic or numerical	C1
	0.17 A	A1
6(b)	(if in neutral) rest of circuit / lamps / television not live / still high voltage / still 240 V or live has high voltage / 240 V / can cause shock	C1
	(if in neutral) <u>when</u> (switched) <u>off</u> circuit / lamps / television not live / still high voltage / still 240 V	A1

- 7.
- (a) (i) supplies the (mains) e.m.f. / voltage B1
- (ii) to complete the circuit / is at 0 V B1
- (b) (i) the circuit / supply is cut / broken **or** current stops B1
fuse melts / blows / burns B1
- (ii) live wire B1
when it cuts the circuit / melts no part of the appliance is live / no shock B1
- 8.
- (a) $(P =)VI$ **or** 230×27 C1
6200 / 6210 W **or** 6.21 / 6.2 kW A1
- (b) (i) 1.1 / 1.12 / 1.1178×10^7 J B1
- (ii) $6.21 \times 0.5 \times 23$ **or** $6.21 \times 30 \times 23$ **or** $3.1 / 3.105 \times 23$ C1
71 / 71.3 / 71.4 / 71.415 c **or** \$ 0.71 / 0.714 / 0.71415 A1
- 9.
- (a) negative charge moves from hair / person / head to balloon C1
electrons move from hair / person / head to the balloon A1
- (b) hair is positive (at end) B1
opposite charges attract B1
or positive and negative attract
- (c) **charges / electrons** B1
- don't flow away
 - aren't conducted (to earth / person)
- stay on balloon / on insulator

- 10.
- (a) to provide a complete circuit (with live) B1
 or to pass current back to mains
 or provide a return path for the current
- (b) current / charge / electrons flow to earth / earth wire / ground (when live touches case) B1
 fuse melts / blows **and** disconnects circuit / cuts live / stops current B1
- (c) doubly insulated B1
 or **case / body** made of plastic / insulator / not made of metal
 or user cannot touch metal
- (d) (circuit breaker) B1
- turns off / acts fast(er)
 - can be reset
 - easy to see it has tripped / switched
 - can detect small difference between live and neutral currents / small (leakage) current to earth
- 11.
- (a) (i) $(I =)P/V$ or 9.6/240 or 9600 C1
 9600/240 or 0.040 C1
 40 A A1
- (ii) any whole number from 41 to 99 (incl.) **with unit** (A) B1
 (e.c.f. from 0.040 A: 1,2,3 A)
- (b) $9.6 \times 25 \times 21$ or $9.6 \times 25/60$ or $9.6 \times 25/60 \times 21$ or 5040 c or \$50.40 etc. C1
 84 c or \$0.84 or €0.84 or £0.84 or Rs0.84 etc. (85.7/86c from 0.42h) A1

12.

- (a) (i) fuse symbol correct B1
in live wire before junction of two elements B1
- (ii) the (metal) case/outside B1
- (iii) 1. live wire touches case; live touches person B1
2. current goes to earth; current does not go through the person B1
fuse blows B1
- (b) (i) most of the energy output is useful/heat; little energy is wasted; B1
- (ii) hot air rises (**not** heat rises) B1
density of hot air is lower B1
convection current mentioned OR hot air rises and cold air falls B1
- (c) (i) 1500 W B1
- (ii) 1. conversion to kW seen on any power; 2.1 (kW) seen C1
5.25; 5.2; 5.3 (kWh) A1
2. $E = P \times t$ in any form, algebraic or using any power or time e.g. 600×2.5 ,
 600×150 C1
 1.89×10^7 (J) OR $3.6 \times 10^6 \times$ (c)(ii)1. A1

13.

- (a) (i)
- | | | | |
|--------|--------|--|--|
| | | | |
| | | | |
| closed | open | | |
| closed | closed | | |
- B1
- B1
- (ii) S_1 closed $\rightarrow$ motor on B1
 S_1 open $\rightarrow$ heater off B1
- (iii) the heater would overheat/burn/melt **or**
more efficient cooking/circulation described B1

(b) (i)	$(I =) P/V$ or 3700/230 or 3500/230 or 200/230 or 15(.217) 16(.08695)A or 16.1 A	C1 A1
(ii)	integral value: 17 $\rightarrow$ 40 A or up to candidate's (b)(i) + 24 live	B1 B1
(iii)	if case becomes live or if live wire touches case fuse blows live/supply disconnected/case safe to touch/prevents shock/ prevented electrocution	B1 B1 B1
(c)	0.20/3.5/3.7 (kW) or 200/3500/3700 $\times$ 12 $\times$ 35 0.20/3.5/3.7 $\times$ 12 $\times$ 35 or 1470 c or 1554 c or 84000 c 84 c or \$0.84 (allow €/£/R etc.)	C1 C1 A1
14.		
(a)	no shock // no electrocution (if) case becomes live // live touches case	B1 B1
(b)	correct conversion to kW, 0.5 seen // conversion to hours // 0.75 // $\frac{45}{60}$ // $(E =) P \times t$ 0.375 // 0.38 // 0.37 (kW h)	C1 A1
15.		
(a) (i)	$(I =)P/V$ or $P = VI$ or 650/230 2.8 or 2.83 A	C1 A1
(ii)	3, 4, 5, 6 or 7 A only	B1
(b) (i)	casing becomes live/at high voltage current through user/user electrocuted/user shocked	B1 B1
(ii)	fuse blows/melts/breaks fuse in live wire/(microwave) disconnected/circuit broken/no current	B1 B1

16.

- (a) (i) high voltage/where voltage (**not** current) arrives/dangerous (wire) B1
- (ii) zero voltage/safe wire B1
- (iii) zero voltage / connected to ground B1
- (b) (i) (wire) heats up/current increases/electrons move faster C1
- (ii) (wire) melts/causes fire (**not** blows/melts fuse) A1
- (c) avoids electrocution/current through person/water is a conductor B1

17.

- (a) 3024 B1
- 3.024 (or 1/1000 of previous answer) B1
- 1.512 (or 1/2 of previous answer) B1
- (b) smaller resistance **accept** more current B1
- (c) heater uses more than 3A **accept** current 12.6A B1
- causes fuse to melt/blow/burn/break B1